Question 1	a	b	c	d	e	f
Question 2	a	b	c	d	e	f
Question 3	a	b	c	d	e	f
Question 4	a	b	c	d	e	f
Question 5	a	b	c	d	e	f
Question 6	a	b	c	d	e	f
Question 7	a	b	c	d	e	f
~~Question 8~~	a	b	c	d	e	f
Question 9	a	b	c	d	e	f
Question 10	a	b	c	d	e	f
Question 11	a	b	c	d	e	f
Question 12	a	b	c	d	e	f
Question 13	a	b	c	d	e	f
Question 14	a	b	c	d	e	f
Question 15	a	b	c	d	e	f
Question 16	a	b	c	d	e	f
Question 17	a	b	c	d	e	f
Question 18	a	b	c	d	e	f
Question 19	a	b	c	d	e	f
Question 20	a	b	c	d	e	f
Question 21	a	b	c	d	e	f
Question 22	a	b	c	d	e	f
Question 23	a	b	c	d	e	f
Question 24	a	b	c	d	e	f
Question 25	a	b	c	d	e	f
Question 26	a	b	c	d	e	f

Solution to problem 1.

a.

At the lower flat position, the substance is melting.

The melting point is $\approx -100\ °C - 50\ °C\,(\frac{3\,cm}{9\,cm}) = -117\ °C$

b.

The boiling position is the upper flat part.

It is $\approx 50\ °C + (\frac{5\,cm}{9\,cm})\ 50\ °C = 77.8\ °C$

c.

For the first 40 seconds all the power is going into converting solid to liquid and after 40 seconds all the solid has melted.

Therefore, the initial mass of the solid , $m = \frac{P.\ \Delta t}{L}$

We need to find P, the constant power.

During the liquid phase:

$dQ = m.\ c_l.\ dT$

$dQ = P.\ dt$

$P.\ dt = m.\ c_l.\ dT$

So we have, $P = (\frac{dT}{dt}).\ m.\ c_l$

From the graph,

$\frac{dT}{dt} = \frac{50\ °C}{60\ s} = 0.83\ °C/s$,

$m = 0.25\ g$, $c = 2.43\ J/g.K$

$P = 0.83\,\frac{K}{s} \times 0.25\ g \times 2.43\ J/g.K = 0.504\ J/s$

Therefore, $m = \frac{0.504\,\frac{J}{s} \times 40\ s}{105\ J/g} = 0.192\ g$

d.

Let L_e be latent heat of evaporation.

so, $m.L_e = P.\Delta t_{evap}$

Where $\Delta t_{evaporation} \approx 700\ s - 280\ s = 420\ s$

So, $L_e = \frac{0.504\frac{J}{s} \times 420s}{0.25 \times 10{-}3\ Kg} = 847\ J/g$

e.

Let c_g be specific heat of gaseous

Like before,

$P = (\frac{dT}{dt}).\ m.\ c_g$

$(\frac{dT}{dt}) = \frac{(200-100)°C}{(820-720)s} = 1\ °C/s$

So, $C_g = \frac{0.504\ J/s}{1\frac{K}{s} \times 0.25\ g} = 2.02\ \frac{J}{gK}$

f.

The ideal gas law is written a: PV=nRT

P is the pressure of the gas in Pascals

V is the volume of the gas in cubic metres

,n is the number of moles

T in kelvins (the Kelvin scale is a shifted Celsius scale, where 0.00 K = −273.15 °C, the lowest possible temperature).

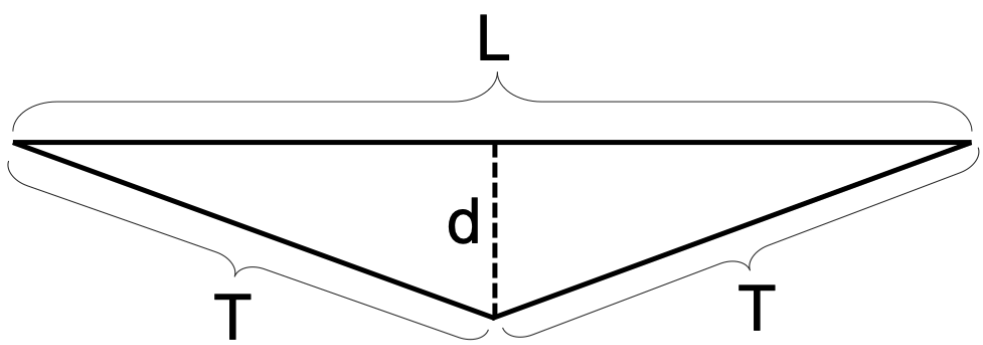
R has the value 8.314 J/(K·mol)

$n = 0.25\ g \times \frac{1\ mol}{46\ g} = 5.435 \times 10^{-3}\ mol$

So, $P = 5.435 \times 10^{-3} \times 8.314\ (200+273.15) / 5 \times 10^{-3} = 4274\ Pa$

Solution to problem 2.

a.



Distance from the ceiling to the center of the ball=d

The change in length: $\Delta l = 2\sqrt{d^2 + (\frac{L}{2})^2} - L$

So, the tension is, $T = K.\ \Delta l$

And the magnitude of the upward force from the tension is: $2T\,(\frac{d}{\frac{L}{2}})$

$$2(K(2\sqrt{d^2 + \left(\frac{L}{2}\right)^2} - L))(\frac{d}{\frac{L}{2}}) = mg$$

$$\frac{4kd}{L}(2\sqrt{d^2 + (\frac{L}{2})^2} - L) = mg$$

Let's define, w=L/2

Then, $\frac{4kd}{2w}(2\sqrt{d^2 + w^2} - 2w) = mg$

$$\frac{4kd}{w}(\sqrt{1 + \left(\frac{d}{w}\right)^2} - 1) = mg$$

Let's define, x = d/w

Then, $4kwx(\sqrt{1 + (x)^2} - 1) = mg$

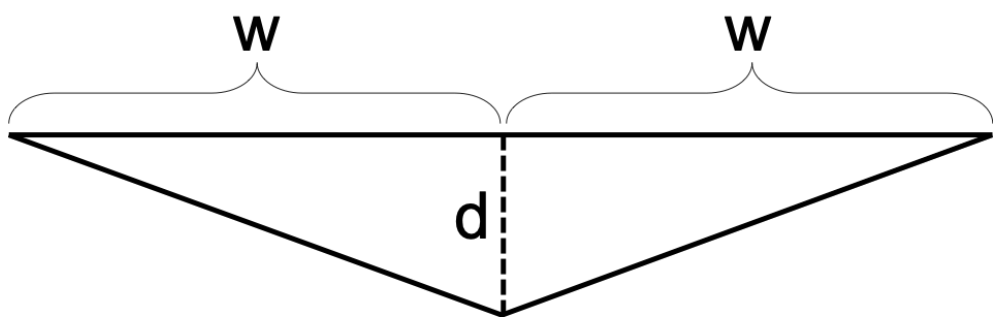
$x(\sqrt{1 + (x)^2} - 1) = mg/2kL$

We can solve it approximately,

$\sqrt{1+\alpha} = 1+\alpha/2$. $\alpha << 1$ so, $x(\sqrt{1+(x)^2}-1)=1/2x^3$, $x \approx 0.163$

Therefore, $d=L/2\times 0.163= 0.122$ *m*

b.



$$m\ddot{y} = -mg + 2K\left(2\sqrt{(d-y)^2+w^2}-2w\right)\left(\frac{d-y}{w}\right)$$

$$= -mg + 2K\left(2\sqrt{d^2+w^2-2yd}-2w\right)\left(\frac{d-y}{w}\right)$$

$$= -mg + 2K\left(2\sqrt{d^2+w^2}\left(1-\frac{yd}{d^2+w^2}\right)-2w\right)\left(\frac{d-y}{w}\right)$$

$$= -mg + 2K\left(2\sqrt{d^2+w^2}-2w\right)-2\frac{yd}{\sqrt{d^2+w^2}}\left(\frac{d}{w}-\frac{y}{w}\right)$$

We defined, $2K\left(2\sqrt{d^2+w^2}-2w\right)\left(\frac{d}{w}\right) = mg$ *, so,*

$$m\ddot{y} = \frac{-4Kyd}{\sqrt{d^2+w^2}}\frac{d}{w} - 2K\left(2\sqrt{d^2+w^2}-2w\right)\frac{y}{w}$$

$$2K\left(2\sqrt{d^2+w^2}-2w\right)\frac{y}{w} = mg\frac{y}{d}$$

$$m\ddot{y} = -4k\frac{d^2}{w\sqrt{d^2+w^2}}y - mgd\frac{y}{d}$$

$$\ddot{y} = -\left(\frac{4kd^2}{mw\sqrt{d^2+w^2}}+\frac{g}{d}\right)y$$

$$\text{so, } T = \frac{2\pi}{\sqrt{\frac{4kd^2}{mw\sqrt{d^2+w^2}}+\frac{g}{d}}} = 0.408\ s$$

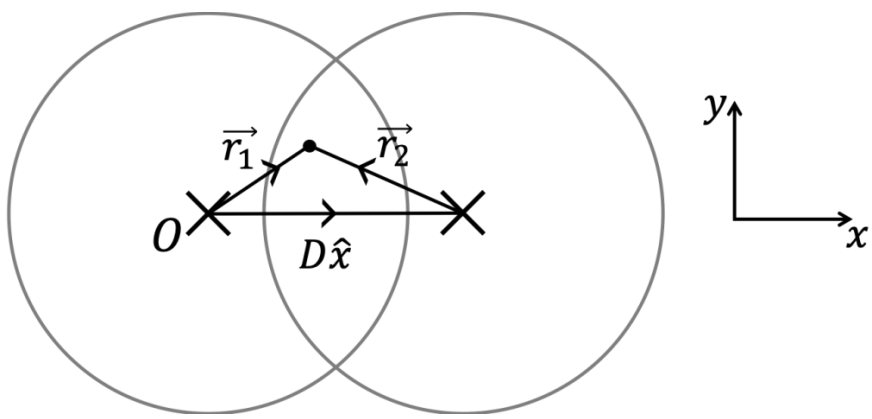
c.

No, it will not be described exactly by simple harmonic motion because simple harmonic motion means:

$\ddot{Y}$= - w^2 y for a constant w and the differential equation above does not have that form only have that form for very small oscillations.

Solution to problem 3.

a.



Put the origin to the center of the left sphere.

Consider a spatial point at $\vec{r_1}$ in the overlapping region.

Let $\vec{r_2}$ be the displacement of that spatial point to center of the right sphere.

Electric field due to the left sphere

$$\vec{E_1} = \frac{1}{4\pi\epsilon_0}\left(\frac{4}{3}\pi|\vec{r_1}|^3\rho_0\right)\frac{\hat{r_1}}{|\vec{r_1}|^2} = \frac{1}{3\epsilon_0}|\vec{r_1}|^3\rho_0\frac{\vec{r_1}}{|\vec{r_1}|^3} = \frac{1}{3\epsilon_0}\rho_0\vec{r_1}$$

Similarly, the electric field due to the right sphere

$$\vec{E_2} = -\frac{1}{3\epsilon_0}\rho_0\vec{r_2} = -\frac{1}{3\epsilon_0}\rho_0(\vec{r_1} - D\hat{x})$$

Total electric fiel: $\vec{E} = \vec{E_1} + \vec{E_2} = \frac{1}{3\epsilon_0}\rho_0\vec{r_1} - \frac{1}{3\epsilon_0}\rho_0(\vec{r_1} - D\hat{x}) = \frac{\rho_0}{3\epsilon_0}D\hat{x}$

b.

Recall the surface bound charge is just the polarization perpendicular to the surface.

In this limit, we can see the two overlapping spheres of opposite charges as a dipole at the center with dipole moment $-(\frac{4}{3}\pi R^3)\rho_0 D\hat{x}$

=> *Polarization* = $-\frac{(\frac{4}{3}\pi R^3)\rho_0 D}{\frac{4}{3}\pi R^3}\hat{x} = -\rho_0 D\hat{x}$

=> *Surface bound charge* $\sigma = -\rho_0 D\hat{x}\cdot\hat{r} = -\rho_0 D cos\theta$

c.

Inside the conducting sphere, electric field is 0.

So the setting of the question is the same having zero electric field inside a sphere and certain constant electric field E_0 *outside the sphere.*

Without lost of generality, we can orient the axis such that the electric field E_0 *is along the x direction.*

This is equivalent to the superposition of the following two settings:
(1) having constant electric field E_0 *both inside and outside the sphere; and*
(2) having constant electric field $-E_0$ *inside the sphere and zero electric field outside the sphere.*

Recall the surface bound charge is given by $\frac{\sigma}{\epsilon_0} = E_{\perp,up} - E_{\perp,down}$ *about the surface*

So setting (1) does not give any bound charge since both $E_{\perp,up}$, $E_{\perp,down}$ *are the same.*
Setting (2) is just the same situation that we had in part b above,
with the electric inside the sphere $-E_0 = \frac{\rho_0}{3\epsilon_0}D$ => $\rho_0 D = -3\epsilon_0 E_0$.
Putting into the result in part b, $\sigma = -\rho_0 D cos\theta = 3\epsilon_0 E_0 cos\theta$

Therefore, the surface bound charge is $3\epsilon_0 E_0 cos\theta$.